

Can't Nobody Stop Me! Non-Euclidean Portal Reset for Continuous Walking in Virtual Reality

Ho Jung Lee , Taewoo Jo , Sulim Chun , and In-Kwon Lee 

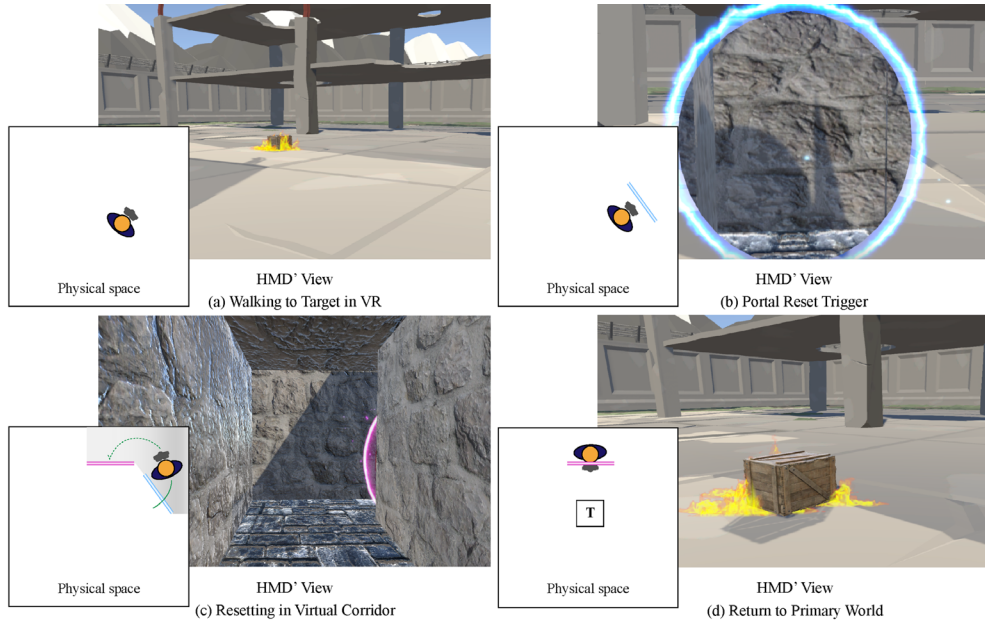


Fig. 1: Illustration of the Non-Euclidean Portal Reset (NEPR) process. (a) A user explores the virtual environment while walking in physical space. The inset shows the user in physical space and the user's HMD view. (b) As the user moves toward a target, such as a burning box, a portal is generated before colliding with the physical space boundary. The blue line represents the portal relative to physical space. (c) Upon entering the portal, the user is transported to a secondary virtual environment, designed as a corridor where the reset occurs. The purple line marks the portal leading back to the primary world, while the gray area represents the virtual corridor. The virtual corridor is dynamically generated to ensure collision-free walking and, by its end, resets the user to a safe heading and position. (d) After completing the reset, the user returns to the primary world through the portal. Users are returned to a location near the target object, ensuring seamless navigation. The inset shows the position of the target object in physical space, denoted by the letter T. Users can take advantage of the NEPR reset, which allows them to reach distant targets without performing multiple resets.

Abstract— Redirected walking (RDW) allows people to explore large virtual environments while walking within a smaller physical space. When physical space is limited, explicit resets such as turn-in-place are unavoidable. Previous studies have reduced the frequency of resets by adjusting user paths. However, resets remain necessary under severe spatial constraints. The frequent use of turn-in-place resets interrupts locomotion and can degrade task performance and user experience. We present Non-Euclidean Portal Reset (NEPR), a reset technique that enables continuous experiences without pausing the user's walking. When a collision risk is detected, NEPR opens a virtual portal leading to a short, non-Euclidean corridor. Traversing the corridor repositions and reorients the user. The exit returns the user near the point of interest target in the primary world, maintaining flow. To evaluate the effectiveness of NEPR, we conducted user experiments comparing (1) the conventional turn-in-place reset, (2) NEPR, and (3) a hybrid method combining both approaches. Our results demonstrate that NEPR and the combined technique significantly improve the user experience and task performance compared to the traditional method. Overall, NEPR reframes resets as seamless transitions rather than interruptions, enhancing the practicality of RDW.

Index Terms—Virtual reality, Redirected walking, Resetting, User Experience

1 INTRODUCTION

Navigation is a key factor in the quality of a Virtual Reality (VR) experience. To support efficient navigation and provide a quality user experience, various locomotion techniques have been studied [29, 49]. Among these techniques, physical walking is regarded as one of the most natural and intuitive approaches. However, the finite size of the tracked physical space necessitates techniques that keep users within safe boundaries [22, 63]. Razzaque et al. [50] introduced Redirected Walking (RDW) as a prominent solution. RDW manipulates the mapping between physical and virtual movements through subtle visual adjustments and adaptive environment modifications, thereby guiding users without their awareness. Although subtle techniques have im-

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proved [3, 21, 40], they still cannot guarantee collision avoidance. Thus, explicit reset techniques that reorient the user's direction are necessary in practice [19, 36, 44, 59].

Williams et al. [71] proposed several methods for resetting. One method is the turn-in-place technique, which is a 2:1 turn that uses rotation gain to rotate 180 degrees in the physical environment and 360 degrees in the virtual environment without freezing the screen. The turn-in-place is a simple and commonly adopted reset baseline that reorients the user in place while keeping the position fixed [1, 41]. Since then, turn-in-place has evolved into various methods to provide an optimal reset direction [3, 32, 62, 78]. These methods have been combined with the redirection gains of RDW, which is currently the most widely used method and allows for a more natural gait [40, 50, 59]. However, current turn-in-place methods apply rotation gains beyond user detection thresholds, which can disrupt the user experience [5, 34, 77]. Furthermore, frequent resets could cause users to walk less and stop more often during the same period of time, which could negatively impact the task performance, such as completion time [6, 59]. Reset and redirection techniques have recently improved to reduce the frequency of resets; however, resets remain an unavoidable problem in VR walking experiences [19, 36, 44]. These limitations are especially evident in goal-directed tasks, such as walking to distant targets, which require repeated resets and stopping [34]. Meanwhile, Flexible Space [68] reconfigures an indoor virtual environment by relocating rooms and rerouting corridors during moments of change blindness. The technique enables users to walk continuously within a tracked area to reach a goal without stopping. However, it is only applicable to indoor virtual environments, and the indoor rooms must be smaller than the physical space being tracked. Inspired by the concept of Flexible Space, we are exploring the possibility of incorporating impossible spatial layout reconfiguration into the reset process. This would overcome the limitations of traditional turn-in-place techniques, which require frequent pauses, and enable continuous walking for task-oriented activities in constrained physical spaces. Accordingly, this study addresses the following research questions:

- **RQ1.** How can spatial layout reconfiguration be used as a reset technique to support non-stop forward walking?
- **RQ2.** Does the proposed method improve task performance and user experience compared to turn-in-place techniques?

To address these issues, we propose the Non-Euclidean Portal Reset (NEPR) system, which uses spatial reconfiguration. As a user approaches a boundary of physical space, NEPR generates a portal in their field of view that leads to a secondary world. This secondary world contains a short, dynamically generated, non-Euclidean corridor according to the user's current physical position and direction. The corridor then guides the user away from the physical boundary. After passing through the exit portal at the end of the corridor, the user returns to the primary world near the navigation target. This process reduces the risk of collision by keeping the user within the boundaries and eliminates in-place rotation that utilizes rotation gain beyond the detection threshold. By exploiting non-Euclidean geometry, the corridor appears short while mapping to a longer displacement in the virtual environment. This effectively advances the user toward the target in goal-directed tasks. Fig. 1 shows the NEPR operation process. We conducted a user study to determine the impact of a traditional turn-in-place reset, NEPR, and a hybrid approach combining both techniques on the VR experience. Our experimental results show that NEPR improves user experience and task performance compared to traditional turn-in-place and that combining turn-in-place and NEPR further improves user experience. The main contributions of this research are as follows:

1. We propose a portal-based, goal-directed reset mechanism (NEPR) that eliminates in-place rotation while preserving the continuity of forward walking.
2. Our experiments confirm that combining NEPR with turn-in-place further enhances the user experience in VR.

2 RELATED WORK

2.1 Redirected Walking

VR spaces can be arbitrarily large, but the tracked physical space rarely matches them, so effective locomotion techniques are essential [29, 43, 49]. Teleportation is widely deployed and supports efficient navigation [8, 70], but the instantaneous viewpoint jump breaks spatial continuity and can reduce presence [7]. In contrast, physical walking-based locomotion, which interacts with the physical space, is the most realistic way of walking, maximizing presence [63] and enabling efficient navigation [52, 58]. However, physical walking has the limitation that it cannot fully cover an infinite virtual space due to the constraints of physical space. To address this issue, Razzaque et al. [50] proposed the Redirected Walking (RDW) technique.

RDW involves subtle manipulation of the user's HMD screen (e.g., subtle technique) or alteration of the structure of the virtual environment (e.g., change blindness) to allow for more extensive exploration of the virtual space within a constrained physical space. Upon detecting the application of redirection gain, users may experience cybersickness or reduced presence. Detection thresholds were examined in various situations where users felt that no redirection gain was being applied, and the measured detection thresholds provided a range of possible gains for RDW operation in subsequent studies [12, 28, 31, 35, 56, 57, 73]. Since then, RDW techniques have evolved by applying different strategies to optimally guide the user in the right direction with redirection gains within a detection threshold [19, 36, 40, 44]. Furthermore, extensive research has been conducted to apply RDW to irregular physical spaces [33, 62, 72, 74]. Nevertheless, when the mismatch between virtual and physical spaces is severe, avoiding collisions still requires interrupting locomotion with explicit resets.

2.2 Reset Techniques

When exploring VR environments using physical walking, the reset process is essential due to the spatial difference between the physical and virtual environments. Even with the use of subtle redirection techniques, the need for resets cannot be completely excluded, and research is ongoing to minimize the frequency of resets [33, 62, 72, 74, 75]. Williams et al. [71] proposed the '2:1-Turn' technique, which applies a rotation gain when performing a reset without freezing the screen, and displays a 'Turn in Place' prompt in the user's field of view. They also suggested the 'Freeze Turn' technique, which rotates the user 180 degrees in place while freezing the screen, and the 'Freeze Backup' technique, which repositions and reorients the user. A reset technique that reorients the user's direction using attention-grabbing distractors has also been studied [46–48]. Williams et al. [73] showed that a distractor broadened the detection threshold for rotation gain, confirming another advantage of distractor as the reset technique. Among various reset techniques, turn-in-place methods like 2:1-turn are frequently utilized in other VR walking studies due to their ease of implementation and rapid readjustment capability [1, 41].

Several methods, such as Modified Reset-to-Center (MR2C), Reset-to-Gradient (R2G), and Step-Forward Reset-to-Gradient (SFR2G), have optimized turn-in-place resets considering irregular physical spaces [3, 62]. Zhang et al. [78] proposed a technique to optimize the reset direction by considering the redirection technique and the reset technique in combination in irregular physical space. Methods that allow users to reposition themselves during reset have also been studied [77]. Lee et al. [32] proposed a technique to provide the optimal reset direction using a reinforcement learning based reseter in a multi-user RDW scenario where multiple users experience the VR environment simultaneously. In addition to optimization studies, a multisensory reset UI has been investigated to direct resets so that users can fully utilize the power of the reseter [34]. Despite these optimizations, interacting with distant virtual targets in limited physical space still often requires multiple resets. In addition, existing reset techniques suffer from continuity problems, requiring the user to stop walking and perform a readjustment. To address these issues, a new approach is needed that can minimize the frequency of resets while maintaining the continuity of the user's movement during the reset process.

2.3 Non-Euclidean based Techniques

In computer graphics and VR, the concept of non-Euclidean geometry is often used to describe spatial structures that contradict the physical consistency of Euclidean geometry. While strictly mathematical non-Euclidean geometry refers to curved spaces rejecting the parallel postulate (e.g., hyperbolic geometry) [23], in a broader VR context, it involves manipulating topology and metric to connect distant points or overlap spaces. Significant research has explored the 3D modeling of such spaces [11, 45], and these geometric structures have been particularly instrumental in VR locomotion research to enable natural walking within limited physical spaces [17, 18, 24, 69].

Specifically, impossible spaces have enabled continuous walking without explicit resets by exploiting change blindness to rearrange connections [60] and by using self-overlap to reuse the same tracked physical space [61]. Flexible Spaces realized virtually infinite walking in room scale via procedurally generated overlapping layouts [68], and overlap-detection studies showed that path complexity reduces detection rates [65]. Techniques to improve the spatial-compression efficiency of self-overlapping environments have also been proposed [66, 67]. Lochner et al. [42] verified whether natural walking utilizing impossible space with multiple overlapping environments through a portal could be a practical alternative to other locomotion methods (teleport, joystick, arm swing). However, when the experienced room exceeds the tracked physical space or in open outdoor scenes with few occluders, explicit resets remain necessary. Motivated by Flexible Spaces, which connects distinct rooms via corridors to afford seemingly infinite walking without stops, we propose a brief portal-based reset that replaces turn-in-place while preserving the percept of continuous locomotion.

Portal-based techniques are another non-Euclidean approach in VR, applying topological shortcuts that connect locations independently of the real path. Portals have been shown to support efficient wayfinding [10] and are preferred for their immersion and continuity among connection/transition techniques [26]. They have diversified to enhance interaction and scale: “view-through” and targetable portals enable manipulation of distant objects with good usability and perceived quality [2]; automatic placement improves navigation efficiency in large environments [41]; and portal-driven exploitation of spatial connectivity yields natural, engaging tours and training scenarios [51]. Cognitive impacts appear acceptable, with portal travel showing no significantly worse memory outcomes than walking or teleportation [64]. Beyond user experience, portals have also been used for resets: a CAVE-based portal reset demonstrated feasibility but required users to select exits and did not preserve continuous gait [14, 21], and redirected teleportation gradually adjusted its direction by using a portal to make more effective use of the tracking space, but it required repeated work and user-generated portals, and could not completely replace turn-in-place resets at boundaries of the tracking space. [39]. To the best of our knowledge, portal-based methods are not yet validated as a general substitute for turn-in-place resets; we address this gap with an automatically scheduled, brief portal-based reset that preserves continuous locomotion.

3 METHOD

3.1 Design Overview

We introduce NEPR, a reset that preserves continuous forward walking by replacing TiP with a brief portal-corridor traversal. NEPR targets limitations of turn-in-place and portal-based resets. Turn-in-place resets typically apply rotation gains beyond the user’s detection threshold to induce reorientation [32, 62, 71, 78]. In constrained tracking spaces, walking interactions frequently push users to the boundary, leading to repeated resets and pauses that can degrade experience and task performance [34, 59]. Inspired by Flexible Spaces [68], where two spaces are connected by a corridor to enable infinite walking, NEPR operates instead when the user approaches the physical boundary: it opens an In-Portal at the impending collision point and places an Exit-Portal near the point of interest (PoI) target the user was walking toward. A non-Euclidean corridor generated by NEPR connects these two points.

This allows the user to continue walking forward without turn-in-place while being repositioned to a location adjacent to the PoI. Previously, portal-based resets often required manual activation or multiple attempts to avoid collisions [21, 39], NEPR instead analyzes user motion in real time and spawns portals aligned with the user’s current walking direction and task context, so resets feel like uninterrupted exploration. The operation process of NEPR is shown in Fig. 1.

We define the primary world as the main VR scene and the secondary world as the space used to readjust position and direction. The portal connecting the two spaces consists of two elements: Entry and Exit [54]. As the user moves toward a target point and approaches the boundary of the physical space, increasing the risk of collision, the In-Portal, which enters the secondary world, is automatically generated within the user’s field of view. The Exit Portal returns the user from the secondary to the primary world. In virtual space, the Exit-Portal is parallel to the In-Portal and positioned a fixed distance from the target along the pre-reset heading. This ensures that the user returns facing toward PoI target. When users pass through the In-Portal generated in the primary world, they are transported to the secondary world, which consists of a virtual corridor, and when they exit through the Out-Portal in the secondary world at the end of the corridor, they return to the Exit-Portal in the primary world. The virtual corridor is designed to avoid collisions with the boundaries of the physical space, and uses the concept of non-Euclidean geometry to connect the portals in a way that would not be possible in the real world, allowing the user to reset naturally as they continue walking. Because the corridor advances the user toward the PoI during the reset, the return places them near the target; this reduces overt resets but adds corridor traversal distance, which we account for in analysis. Fig. 2 and Fig. 3 illustrate the operation of NEPR with top-down views of the physical and virtual spaces, respectively.

3.2 Portal Reset Trigger

NEPR is a way to reset a user when they are in danger of colliding with the physical environment (PE) by creating a portal in the user’s field of view and allowing them to enter a secondary world. To achieve this, it continuously checks whether the portal reset needs to be activated (Trigger) while the user is walking in real time. Let the unit vector representing the user’s walking direction be θ_p and the user’s position in physical space be p . The center point of the In-Portal (C_{in}) is expressed as follows:

$$C_{in} = p + d_p \theta_p, \quad (1)$$

where d_p is the distance between the user’s current position and the center point of the portal. Defining the width of the portal to be W , the left and right corners of the portal are placed perpendicular to the walking direction θ_p , in other words, along $\theta_p^\perp$, each separated by $W/2$, which is defined as follows:

$$P_l = C_{in} + \frac{W}{2} \theta_p^\perp, \quad P_r = C_{in} - \frac{W}{2} \theta_p^\perp, \quad (2)$$

where P_l and P_r denote the positions of the left and right corners of the portal, respectively. We define the minimum vertical distance from the left and right corners of the portal to the boundary of the PE as L and R , respectively, which are computed as follows:

$$\begin{cases} L = \min_{q \in \text{PE}} \{ \|q - P_l\| \mid (q - P_l) \cdot \theta_p^\perp = 0 \}, \\ R = \min_{q \in \text{PE}} \{ \|q - P_r\| \mid (q - P_r) \cdot \theta_p^\perp = 0 \}, \end{cases} \quad (3)$$

where q denotes a point located on the boundary of the PE. After calculating the distances from P_l and P_r to points perpendicular to the PE boundary, the smallest value is selected, which allows for a more precise determination of whether or not to trigger a portal reset. The Trigger value, which uses the calculated L and R to determine whether to trigger a portal reset, is defined as follows:

$$\text{Trigger} = \begin{cases} 1, & \text{if } (R < 0.75m \wedge L < 0.75m), \\ 1, & \text{else if } (R < 0.5m \vee L < 0.5m), \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

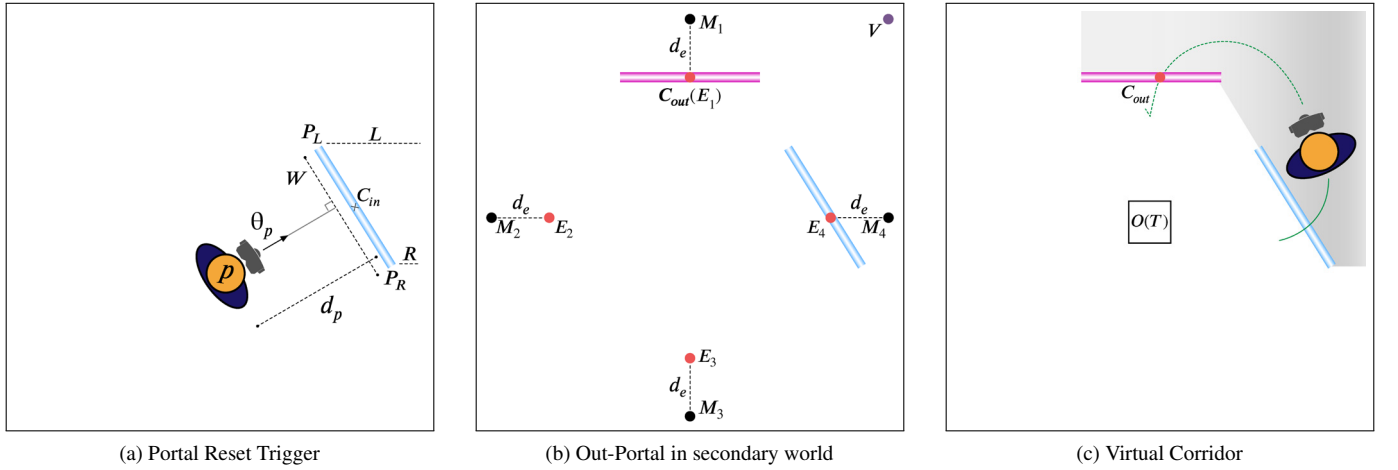


Fig. 2: Illustration of the operation of NEPR from top views in physical space. (a) Showing the situation where a portal reset is triggered. The user is walking at position (p) and moving in direction (θ_p). When the user approaches the boundary of physical space and there is a risk of collision, an In-Portal (blue line, center position; C_{in}) of length W is generated perpendicular to θ_p at a distance d_p from the user's walking direction. The positions of the left and right corners of the portal are denoted by P_L and P_R , respectively. We compute the minimum vertical distances L and R between the physical space and the left and right corners of this portal. We trigger slightly before a predicted boundary violation, ensuring that the secondary corridor beyond the In-Portal is guaranteed W of usable width for safe walking. (b) This is the process of determining the position and direction of the Out-Portal (purple line) that can return to the primary world based on the In-Portal and physical space. Candidates are selected from the center points M_i on each side of the physical space at positions E_i that are vertically far as d_e from the center direction of the physical space. Among the candidates, the candidate that can be reached from the In-Portal by passing through the corner V of the physical space once is finally determined as the position C_{out} of the Out-Portal. The direction of the portal is parallel to the side of the PE boundary used to compute the candidates. (c) Based on the determined In-Portal (C_{in}) and Out-Portal (C_{out}), a virtual corridor is created by connecting the vertices of the physical space boundary (gray polygon). As the user traverses the corridor, they are repositioned upon re-entry to the primary world so that the center of physical space (O) coincides with the original target (T), effectively resetting position and heading.

where Trigger is 1, the portal reset is triggered, and 0, the portal reset is not triggered. If both corners approach within 0.75 meters of the PE boundary, or if either corner approaches 0.5 meters or less, there is a high risk of being blocked from walking after passing through the portal. When these conditions trigger a portal reset, a portal immediately appears in the user's field of view in the primary virtual world. This technique can be used to predict the risk of collision based on the user's location in physical space, and to ensure the minimum width of the virtual corridor in the secondary world that will be traversed beyond the In-Portal. The trigger conditions (0.5m, 0.75m) and parameters ($d_p = 1.0m$, $W = 1.0m$) were determined through pilot tests to balance safety margins with kinematic continuity, ensuring users can walk naturally without clipping virtual walls. Fig. 2a shows the working of this method.

3.3 Generating Secondary World

The secondary world generated by the NEPR technique consists of a virtual corridor along which users walk, readjusting their position and direction in the physical space before returning to the primary world. Immediately after the user passes through the In-Portal, the walls of the virtual corridor corresponding to the same position as the boundary of the physical space appear in the field of view, forming the boundary of the virtual corridor and preventing the user from colliding with the real boundary of the physical space. The virtual corridor is generated so that the user walks clockwise or counterclockwise around the physical-space center, depending on boundary proximity. The relative values of L and R used to calculate the Trigger are used to determine the direction of the corridor. If the value of L is less than the value of R , it means that the left corner of the portal is closer to the boundary of physical space, and the user will perform a reset by walking clockwise with the left area blocked by the wall after passing through the portal. Conversely, if the value of R is smaller, the virtual corridor is configured so that the user can perform a reset by walking counterclockwise.

The Out-Portal is located at the end of the corridor that returns users to the primary world. When determining the Out-Portal location, we

place it to reduce collision risk with the PE boundary during subsequent navigation. Also, the In-Portal (to the secondary world) and the Out-Portal (back to the primary world) should be appropriately positioned so that they are naturally connected by a virtual corridor that preserves continuous walking. Shorter corridors improve efficiency, but overly short ones risk portal overlap or excessive curvature, degrading natural walking. Therefore, in order to solve these problems and create an optimal walking environment, this study defines the criteria for the location of the Out-Portal and designs a secondary world by constructing a virtual corridor that connects to the previously created In-Portal.

The user performs the reset by walking through a virtual corridor that corresponds to the outer area of the physical space. The NEPR should be designed so that, once users have completed the reset, they are guided to an area where they can easily turn left or right and where there is enough space to walk forward. To achieve this, candidate locations for the Out-Portal are placed parallel to the boundary at a distance vertically from the center of each boundary in the physical space. This is designed to facilitate orientation when the user returns to the primary world through the portal, by allowing them to continue their movement relative to the center of each side within the PE. The locations of the portal candidates (E_i) are calculated as follows:

$$E_i = M_i + d_e n_i, \quad i \in \{1, 2, 3, 4\}, \quad (5)$$

where M_i is the center point of the i th boundary of the PE, and n_i is the unit normal vector representing the direction perpendicular to the i th boundary. We set an offset $d_e = 1.0m$ from each PE-edge center, chosen in pilots to guarantee at least W corridor width during reset. From the four candidates, one was chosen that was not too close to the In-Portal to cause portals to overlap or increase the curvature of the corridor excessively, but not so far away as to require unnecessary movement. To achieve this, we determine the center position (C_{out}) of the Out-Portal as the position of the candidate edge E_i reached by traversing exactly one vertex V of the PE boundary in a clockwise or counterclockwise direction, starting from the In-Portal position. The virtual corridor is then constructed as a polygon connecting the left

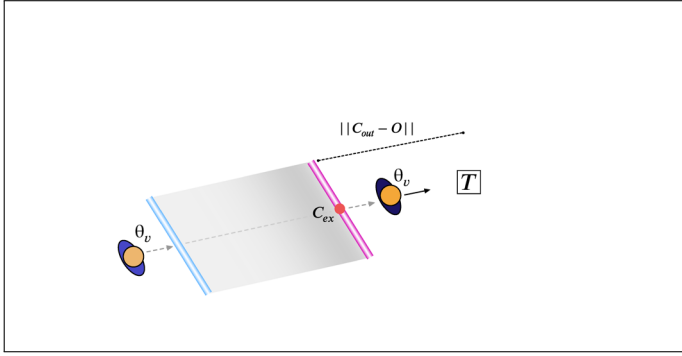


Fig. 3: Illustration showing a top view of the NEPR operation in virtual space. The brightly colored users in the left part of the figure represent the users before the reset, walking in the θ_v . When a portal reset is triggered, an In-Portal (blue line) is generated in front of the user. The dark colored user (right part of the figure) completes the reset by walking through the virtual corridor and exiting from the portal at the Exit-Portal (purple line), indicating that the user has returned to the primary world. The Exit-Portal is placed perpendicular to θ_v at C_{ex} , offset from the target T by $\|C_{out} - O\|$ (the distance between the previous Out-Portal center and the PE center O). The Exit Portal in the primary world is directly connected to the Out Portal that was previously generated in the secondary world.

and right edges of the portals at positions C_{in} and C_{out} , as well as the corresponding edges aligned with the boundaries of the physical space. When users pass through the virtual corridor and return to the primary world, they are repositioned and reoriented so that continuous walking can occur without collisions. This reset process allows for a natural walking without applying any rotation gains. The parameter used in this method is $d_e = 1.0m$. Fig. 2b and Fig. 2c respectively illustrate the procedure for determining the position of the Out-Portal and the resulting configuration of the virtual corridor.

3.4 Return to Primary World

After returning to the primary world, users continue to move toward their previous target (PoI). However, when the distance to the target is far, the risk of colliding with the PE boundary again increases, potentially requiring additional resets, which may negatively impact the experience for users. To avoid this, the location of the Exit-Portal in the primary world should be properly positioned to minimize the number of resets in the user's future walks after the reset. The Exit-Portal is directly connected to the previously determined Out-Portal.

We set up the virtual corridor to lead the user to the C_{out} location relative to the physical space, facing the center of the physical space. As the user explores the virtual space in the primary world, the center of the physical space has the least risk of collision, so this must be considered when placing the Exit-Portal. The position C_{ex} of the Exit-Portal is determined by considering the relationship between the user's target position T , the current user's position C_{out} , and the center of the physical space O . Let θ_v denote the user's orientation before entering the In-Portal, which is perpendicular to the portals placed in the virtual space. The center point of the Exit-Portal (C_{ex}) is calculated as follows:

$$C_{ex} = T - \|C_{out} - O\|\theta_v \quad (6)$$

where T is the virtual-space position of the target the user is moving toward in the primary world. The distance between the target position T and the user's position in the physical space serves as the reference point for the Exit-Portal, and the Exit-Portal is ultimately positioned to lead the user to the PE center with respect to the target. This placement is intended to guide users toward the PE center, and we expect it to reduce subsequent resets, which we will evaluate by measuring the frequency of resets. Experiments were conducted in a $6m \times 6m$ tracking space. Fig. 3 shows how the technique works.

3.5 Combined with Turn-in-Place

NEPR avoids boundary collisions and advances the user toward the target with a single reset while preserving continuous walking, minimizing interruptions. However, NEPR can degrade the user experience when the target is close due to the additional walking distance. Portal transitions may disrupt spatial memory because they are teleporting through a portal [29]. Therefore, we propose a hybrid method that uses a combination of turn-in-place and NEPR depending on the specific case. The hybrid method applies turn-in-place to users when the distance to the target is less than a threshold distance D , and NEPR when the distance is greater than or equal to D . The goal is to minimize the degradation of the user experience and to provide efficient navigation. We set $D = 12m$ based on pilot results indicating this was the efficiency break-even point where the cost of corridor traversal became lower than the cumulative cost of frequent TiP resets.

4 USER STUDY

4.1 Dependent Variables and Hypothesis

We introduce a portal reset that reconfigures the spatial layout at the boundary, allowing the user to continue walking forward. To evaluate its practical value, we conducted a within-subjects comparison with three techniques: Turn-in-Place (TiP) as the baseline, our proposed portal reset (NEPR), and a Hybrid method (Hm) combining both. To address our research questions, we organized the dependent variables into two categories: task performance, derived from logging data, and user experience, obtained from standardized questionnaires. Brief post-study interviews were included to provide qualitative context and preference rankings.

4.1.1 Task Performance

For goal-directed tasks, we test whether our method improves task performance using three objective metrics: Mean Reset between Targets (MRbT), Total Walked Distance (TWD), and Experiment Time (ET). These metrics are widely used in RDW and virtual wayfinding research [7, 15]. MRbT is the average number of overt resets between consecutive targets, reflecting the interruption cost of reorientation; reset counts and distance-between-resets are primary criteria in RDW evaluations because overt resets disrupt locomotion and can reduce flow and presence [32, 62, 71, 78]. TWD summarizes the spatial efficiency of the physical route required to accomplish the same virtual objectives [15, 52]. ET measures the total time cost to complete the target sequence [7]. NEPR reconfigures the spatial layout at the boundary. This technique is related to the methods of impossible and flexible spaces that use spatial overlap to reduce the physical distance to the same virtual objectives [42, 60, 61, 68]. Therefore, in goal-directed completion tasks, We hypothesized that NEPR would reduce MRbT, ET, and TWD relative to TiP, reflecting higher spatial efficiency. We expect the Hm to show the same directional advantages relative to TiP. To this end, we have formulated two research hypotheses, which are as follows:

- **Hypothesis 1 (H1):** NEPR will significantly improve task performance—specifically, by reducing MRbT, ET, and TWD—compared to turn-in-place method.
- **Hypothesis 2 (H2):** Hybrid method will significantly improve task performance—specifically, by reducing MRbT, ET, and TWD—compared to turn-in-place method.

4.1.2 User Experience

We evaluated user experience using standardized questionnaires and interviews. Cybersickness was measured using the Simulator Sickness Questionnaire (SSQ) (4-point categorical scale) [27], while immersion and presence were assessed using the Engagement, Enjoyment, and Immersion questionnaire (E2I) [38] and the Igroup Presence Questionnaire (IPQ) [53] (7-point rating scales). Usability was evaluated via the System Usability Scale (SUS) (5-point scale) [9]. Cognitive load was measured using the NASA Task Load Index (NASA-TLX) [25],

assessing six subscales—mental, physical, and temporal demand, performance, effort, and frustration—on a scale from 0 to 100 with 5-point increments. Finally, participants provided preference rankings and qualitative feedback through semi-structured interviews. Based on this dataset, we aimed to validate whether NEPR and hybrid techniques improve the user experience. For this purpose, we formulated two research hypotheses, which are as follows:

- **Hypothesis 3 (H3):** NEPR will significantly enhance user experience—specifically, by reducing SSQ and NASA-TLX scores and increasing SUS, E2I, and IPQ scores—compared to the Turn-in-Place method.
- **Hypothesis 4 (H4):** Hm will significantly enhance user experience—specifically, by reducing SSQ and NASA-TLX scores and increasing SUS, E2I, and IPQ scores—compared to the Turn-in-Place method.

4.2 Experimental Design

The experiment was conducted in a physical space of $6 \times 6m^2$. We conducted the VR experiment using HTC VIVE Cosmos Elite HMD and placed four base stations in the corners of the physical space to track the participant's orientation. The desktop used in our experiments had the following specs: Intel i7-12700 CPU, NVIDIA RTX 3070 Ti GPU, and 64GB RAM. The HMD device communicates wirelessly with a desktop installed outside the physical space where the experiment is being conducted. The participant also participates in the experiment by holding a controller in each hand. The virtual space that users will experience is realized as a ruined apocalyptic city and is $100 \times 100m^2$ larger than the physical space where the experiment is being conducted. The environment runs at a frame rate of 90fps.

To instantiate a goal-directed navigation task, participants completed a repeating mission: locate a burning supply crate, extinguish it with the left-hand extinguisher, and then collect it with a right-hand grab. After each collection, a new crate was spawned to continue the sequence. Collecting 10 supply crates in total completes one experiment. This process was repeated three times, each time using a different reset method. To ensure the task consistently required approaching explicit goals, targets were placed around each participant at random orientations and distances sampled uniformly from 4 to 20 meters. To ensure fairness across participants, each person received an independently randomized sequence with the same distance distribution, and each block included a balanced mix of near and far targets within the 4–20 meters range, with order randomized per participant. Controllers were rendered as hands in VR, the left hand equipped with a fire extinguisher that emitted powdered gas when the designated button was pressed, and the right hand used a grab action to collect the crate once fully extinguished. Ambient environmental audio (e.g., birdsong and wind) and short interaction sounds (powdered-gas hiss when the extinguisher is used) were included to reduce distraction from real-world noise without altering task demands. When NEPR was activated, a gray-brick virtual corridor appeared, styled to match the primary environment to minimize visual discontinuity. Fig. 1 shows the appearance of the environment used in the experiment.

The experimental environment was built on OpenRDW [37]. One of the experimental conditions, Turn-in-Place (TiP), utilizes the MR2C algorithm [62]. This technique resets the user to the center of the physical space by instructing them to rotate in place. For the TiP condition, we explicitly set the reset trigger buffer distance to $0.5m$. When the reset is triggered, a UI prompts the user with text and an arrow indicating the direction of the turn. The reset concludes, and the UI disappears, once the user completes the required rotation to align with the target direction (i.e., toward the physical center). NEPR, on the other hand, generates a portal in front of the walking user based on the rules defined in the previous section and enters the secondary world to perform the reset. The portal is an ellipse $2.0m$ tall and $1.0m$ wide, and the In-Portal is rendered in blue and the Out-Portal is rendered in purple. Each portal renders a view of the connected primary and secondary worlds. The portals used in the actual experiment are shown in Fig. 1. To isolate reset-mechanism effects, we disabled continuous redirection

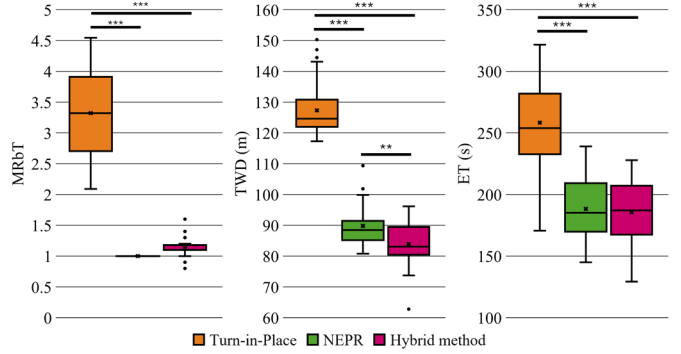


Fig. 4: Means (M) and standard deviations (SD) of the recorded data (MRbT: Mean Resets between Targets, TWD: Total Walked Distance, ET: Experimental Time) for each condition in the user study. Error bars indicate 95% confidence intervals. Statistically significant differences identified by post-hoc tests are indicated by connecting lines, where *, **, and *** represent p-values less than 0.05, 0.01, and 0.001, respectively.

gains and enforced a 1:1 physical–virtual mapping. This ensured that the observed differences resulted from the resetting mechanism. We implemented the experiment in Unity 3D engine (version 2021.2.0f1).

4.3 Participants

A total of 28 participants (male : female = 14 : 14) attended the experiment. When comparing the experimental data using G*Power 3.1 [20], we found that at least 28 participants were needed to meet the conditions of significance level ($= 0.05$), medium effect size ($= 0.25$), and statistical power ($= 0.8$) [13]. We recruited participants through campus-wide advertisements offering \$12 for participation in the experiment. Participants ranged in age from 21–28 ($M = 24.25$, $SD = 1.99$). Nine participants reported having tried VR once or twice, 14 participants had tried VR three or more times, and the remainder had never tried VR. The pre-SSQ survey, administered before the start of the experiment to check for cybersickness, had a mean score of 1.73 ($SD = 11.90$).

4.4 Experimental Procedure

Upon arrival, participants were briefed on the study, provided informed consent, and completed demographics (age, gender, prior VR experience) and a baseline pre-SSQ [27]. After a short tutorial in the HMD to practice each technique, participants performed the same goal-directed task under all three conditions (TiP, NEPR, Hm); the condition order was counterbalanced per participant to mitigate carryover effects. During each trial we logged MRbT, TWD, and ET as objective performance measures, and we recorded gait data continuously. Immediately after each condition, participants completed user-experience questionnaires (SSQ, E2I, IPQ, SUS, NASA-TLX). After all conditions, they provided a preference ranking and completed a brief semi-structured interview about perceived pros, cons, and desired improvements. The session lasted approximately 60 minutes. This protocol enables direct comparison of task performance and user experience across reset techniques while controlling for inter-participant variability. The study was approved by the Institutional Review Board of Yonsei University (No. 7001988-202502-HR-2547-03).

5 RESULTS

The collected data were statistically analyzed to confirm their significance and to determine whether the experimental hypotheses we set were supported. All data were tested for normality by checking that skewness and kurtosis did not exceed 3.0 and 10.0, respectively [30]. We used a one-way repeated measures analysis of variance (RM-ANOVA) test to determine the effects of the reset techniques on the dependent variables. We performed a Greenhouse-Geisser correction

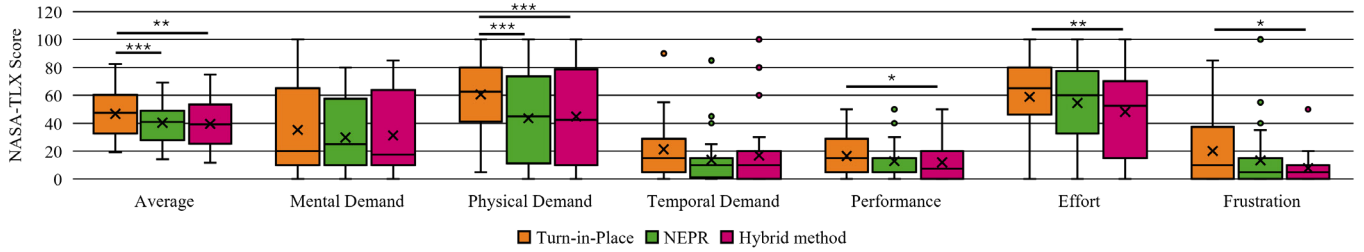


Fig. 5: Means (M) and standard deviations (SD) results of NASA Task Load Index (NASA-TLX) scores for each condition in user study. Error bars indicate 95% confidence intervals. Statistically significant differences identified by post-hoc tests are indicated by connecting lines, where *, **, and *** represent p-values less than 0.05, 0.01, and 0.001, respectively.

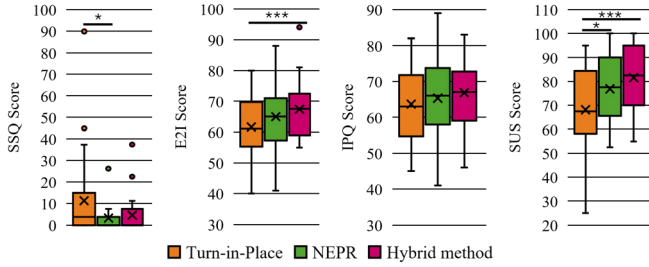


Fig. 6: Means (M) and standard deviations (SD) results of standardized questionnaires (Simulator sickness questionnaire (SSQ), Engagement, enjoyment, and immersion (E2I), Igroup presence questionnaire (IPQ), and System usability scale (SUS)) scores for each condition in user study. Error bars indicate 95% confidence intervals. Statistically significant differences identified by post-hoc tests are indicated by connecting lines, where *, **, and *** represent p-values less than 0.05, 0.01, and 0.001, respectively.

for data that violated the sphericity test. When the results were significant, we performed post-hoc comparisons with Bonferroni correction to determine the differences between conditions.

5.1 Logging Data

We analyzed the logging data to verify the research hypotheses **H1** and **H2**, which were designed to determine if the NEPR and hybrid method improved navigation efficiency over traditional turn-in-place. Fig. 4 shows the results of the logging data. We found that there were significant differences in MRbT ($F(1.06, 28.67) = 282.41, \eta_p^2 = 0.91, p < .001$), TWD ($F(2.00, 54.00) = 458.98, \eta_p^2 = 0.94, p < .001$), and ET ($F(2.00, 54.00) = 177.46, \eta_p^2 = 0.87, p < .001$).

We performed a post-hoc analysis to confirm that TiP in MRbT had significantly more resets than NEPR ($t = 21.15, p < .001$) and Hm ($t = 19.97, p < .001$). In TWD, we found that TiP walked significantly longer distances during the experiment than NEPR ($t = 24.15, p < .001$) and Hm ($t = 27.92, p < .001$). We also found that NEPR walked significantly longer distances than Hm ($t = 3.78, p < .01$). In ET, we found that TiP walked significantly longer than NEPR ($t = 16.00, p < .001$) and Hm ($t = 16.61, p < .001$).

5.2 Standardized Questionnaires

We analyzed standardized questionnaires to test our research hypotheses **H3** and **H4**, assessing whether NEPR and the Hybrid method improve user experience relative to TiP. Fig. 5 and Fig. 6 show the detailed results of NASA-TLX and the results of the questionnaires (SSQ, E2I, IPQ, SUS), respectively. Regarding cybersickness, the ANOVA revealed significant differences in SSQ scores ($F(1.30, 35.17) = 3.94, \eta_p^2 = 0.13, p < .05$). Post-hoc tests indicated that TiP resulted in significantly higher sickness compared to NEPR ($t = 2.62, p < .05$). For immersion, we found significant differences in E2I scores ($F(2.00, 54.00) = 8.21, \eta_p^2 = 0.23, p < .001$). Post-hoc analysis showed that TiP yielded

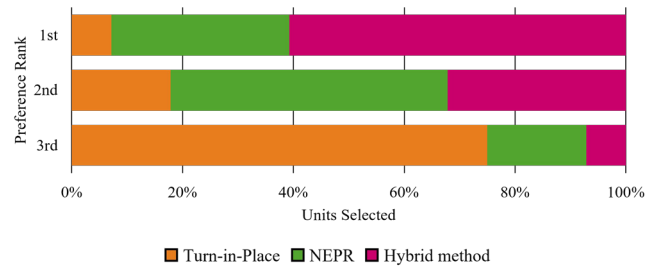


Fig. 7: User preference rankings for each condition, represented by the proportion selected at each rank in the user study

significantly lower immersion than Hm ($t = -4.03, p < .001$). However, IPQ scores did not differ significantly across conditions. In terms of usability, the analysis of SUS scores showed a significant main effect ($F(1.28, 34.66) = 8.82, \eta_p^2 = 0.25, p < .01$). Post-hoc comparisons revealed that TiP had significantly lower usability than both NEPR ($t = -2.69, p < .05$) and Hm ($t = -4.14, p < .001$).

Finally, for cognitive load measured by NASA-TLX, we found significant differences in the overall workload ($F(2.00, 54.00) = 10.22, \eta_p^2 = 0.28, p < .001$). Post-hoc tests showed that TiP induced a significantly higher cognitive load than NEPR ($t = 3.65, p < .01$) and Hm ($t = 4.14, p < .001$). We further analyzed the NASA-TLX sub-scales. Significant differences were found in Physical Demand ($F(2.00, 54.00) = 13.45, \eta_p^2 = 0.33, p < .001$), where TiP scored higher than NEPR ($t = 4.65, p < .001$) and Hm ($t = 4.31, p < .001$). For Performance, there was a significant difference ($F(1.52, 41.10) = 3.95, \eta_p^2 = 0.13, p < .05$), with TiP showing higher load scores than Hm ($t = 2.66, p < .05$). Significant effects were also observed for Effort ($F(2.00, 54.00) = 6.12, \eta_p^2 = 0.19, p < .01$) and Frustration ($F(1.48, 39.94) = 3.84, \eta_p^2 = 0.12, p < .05$); in both cases, TiP scored significantly higher than Hm ($t = 3.48, p < .01$ and $t = 2.77, p < .05$, respectively). No significant differences were found for Mental Demand and Temporal Demand.

5.3 Interview

We analyzed the qualitative differences between the reset techniques by conducting user interviews and comparing their advantages and disadvantages. In the preference ranking, 17 (60.71%) participants preferred Hm, 9 (32.14%) participants preferred NEPR, and 2 (7.15%) participants preferred TiP as their first choice. Fig. 7 shows a stacked bar of participants' stated preferences for each condition.

In the TiP condition, 7 (25.0%) participants found the method intuitive and easy to understand, while 9 (32.1%) reported increased presence due to constant visual engagement with the environment. However, most (22; 78.6%) experienced dizziness and discomfort due to frequent resets when navigating to distant targets. In addition, some felt uncomfortable performing resets directly in front of targets (5; 17.9%), and others found the abruptly generated UI distracting (5; 17.9%). In

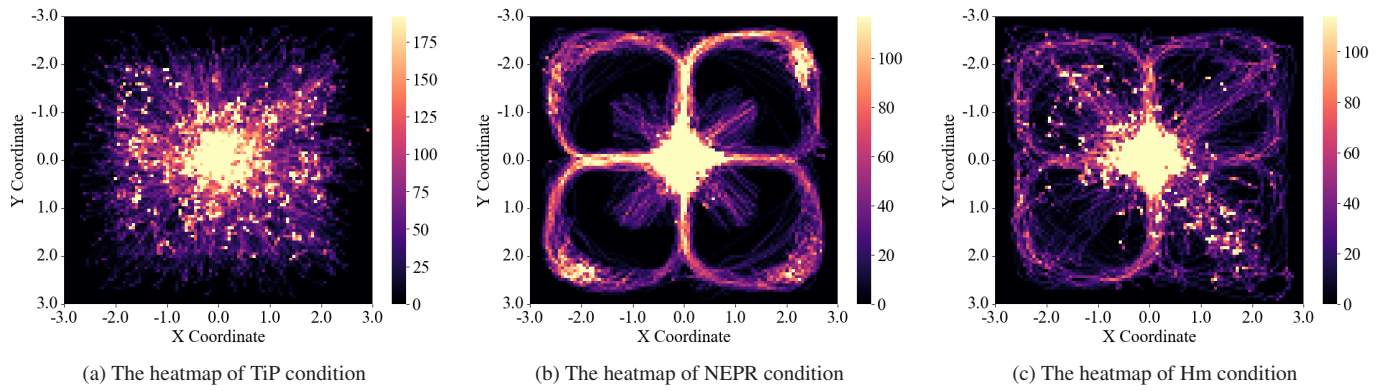


Fig. 8: Heatmaps illustrating the spatial distribution of participants across different conditions. The heatmaps represent the proportion of occupied grids in the physical space, with lighter colors indicating higher occupancy rates and darker colors indicating lower occupancy rates. The x - and y -axes denote the physical space coordinates.

the NEPR condition, more than half (15; 53.6%) appreciated being able to move quickly over long distances to complete their tasks. Participants reported less dizziness and a more realistic walking experience due to continuous movement without turn-in-place (5; 17.9%). Additionally, increased presence due to realistic corridor design (3; 10.7%) and reduced fear of collision due to clearly visible physical boundaries (3; 10.7%) were noted. Conversely, some (4; 14.3%) felt that portals were unrealistic and disrupted presence by introducing different content at each reset. Similar to TiP, some reported inconvenience due to unnecessary walking during resets for nearby targets (7; 25.0%) and blurred spatial memory after resets (4; 14.3%). For the Hm condition, most (20; 71.4%) participants considered it to effectively combine the strengths of both methods, with two (7.1%) specifically noting no dizziness. However, a few (3; 10.7%) found the frequent switching between reset methods confusing.

Regarding the width of the NEPR corridors, most participants (25; 89.3%) found it appropriate, while a few (3; 10.7%) found it too narrow. For the threshold distance ($D = 12m$) used in the Hm condition, the majority (22; 78.6%) found it appropriate, although one (1; 3.6%) said it was too long and several (5; 17.8%) found it too short. In addition, a small number of participants (2; 7.1%) expressed a willingness to require techniques that open the portal intentionally, rather than relying solely on automatic triggering. We implemented dynamic, real-time portal generation; during interviews, participants did not report any delay in portal appearance.

6 DISCUSSION

6.1 Task Performance

Our findings partially support **H1** and **H2**. Relative to TiP, NEPR reduced the reset frequency (MRbT) by converting imminent boundary collisions into forward progress via portal corridors. This is consistent with reports that TiP often requires multiple resets to reach distant targets in constrained spaces [34]. While the expanded trigger buffer (0.5m–0.75m) theoretically limits the navigable space, NEPR offsets this risk by translating users toward the near PoI, thereby reducing the overall reset frequency compared to TiP, which only reorients. However, the extent of this reduction requires further quantitative analysis. Regarding TWD and ET, although NEPR showed reductions compared to TiP, it is important to interpret this not merely as an increase in efficiency, but as a trade-off between physical cost and virtual gain. NEPR inherently incurs a physical cost due to traversing the corridor. However, the embedded displacement toward the target provided by the system effectively substitutes the walking effort required in the primary world, resulting in a net reduction in total distance and time. Although NEPR reduced TWD and ET compared to TiP, it exhibited higher TWD than Hm for nearby targets due to the cost of traversing corridors. Hm, which switches between TiP and NEPR using a distance

threshold D , minimized MRbT as much as NEPR did and achieved the lowest TWD and ET among the three by avoiding corridors for short displacements. While the pilot-chosen D was generally acceptable to participants, performance is likely dependent on the user and task. A brief calibration accounting for rotation tendencies and fatigue could set D to minimize ET under a target MRbT. Consistent with prior work, interviews revealed discomfort when resets occurred directly in front of PoI [76]. In our study, we observed such resets under TiP and NEPR, but not under Hm. Hm's distance-based switching avoided using corridors for short displacements, thereby preventing resets in front of PoIs. Taken together, these findings suggest that NEPR may serve as a viable alternative to TiP for goal-driven navigation tasks in constrained physical spaces.

6.2 User Experience

The results partially support **H3** and **H4**. Compared to TiP, NEPR and Hm produced better subjective results, including lower cybersickness, higher immersion, lower perceived workload, and higher usability. Presence (IPQ) did not differ significantly across techniques. Consistent with prior reports that excessive rotation gains increase cybersickness [59], NEPR's avoidance of large rotation gains was associated with reduced cybersickness compared to TiP. However, because Hm still invokes TiP for short displacements, rotation-related symptoms were not fully eliminated. Nonetheless, by minimizing unnecessary walking and resets through distance-based switching, Hm lowered perceived workload and improved usability. Previous studies have shown that explicit resets can reduce presence by interrupting continuous locomotion [59]. In our study, Hm reduced the frequency and abruptness of resets relative to TiP, which raised perceived immersion in E2I, but presence did not change. This is consistent with Slater's view that presence mainly depends on stable sensorimotor contingencies, while immersion is a system property, since resets still occurred and the display and tracking were identical across conditions [55]. NEPR maintains kinematic continuity during portal resets, unlike TiP, which disrupts gait rhythm. NEPR allows users to maintain their walking speed and proprioceptive flow, helping to mitigate the visual discontinuity of the portal and improve usability scores. Beyond comfort, collision prevention remains a primary objective of reset techniques. NEPR's explicit corridor made physical boundaries legible and reduced collision concerns, in contrast to prior reports that TiP can elicit fear of collision under rotation-based resets [34]. Nevertheless, as revealed in interviews, the visual discontinuity of the secondary world can impact spatial memory. However, the NASA-TLX results showed no significant increase in Mental Demand compared to TiP, suggesting that the cognitive cost of this segmentation did not exceed the threshold to degrade overall task performance. Therefore, future designs should closely match the secondary world to the primary environment to ensure a seamless and coherent user experience.

6.3 Walking Data Analysis

We collected the user's position coordinates in physical space at a frame rate of 90 fps during each experimental condition. Using the data, we visualized the spatial distribution for each condition on a 100×100 grid as heatmaps (Fig. 8). The results show that TiP produced relatively dispersed patterns of movement, with high concentration near the center and progressively lower density toward the boundaries, primarily due to the characteristics of the MR2C resetter, which reset users to the center [62]. In addition, the reset buffer prevented users from approaching within 50cm of boundaries, limiting usable space. Conversely, NEPR produced well-structured, symmetrical patterns with defined looping paths. The heatmap suggests that NEPR effectively guided users along predictable paths, reducing irregular movement and optimizing space utilization. For Hm, density patterns near the center were similar to TiP, but looped path structures similar to NEPR were also observed, suggesting that Hm successfully implemented a hybrid approach combining free exploration and structured movement. Observations of unused regions in NEPR suggest that future research should refine the corridor generation logic to account for obstacles and irregular physical spaces. Furthermore, Hm can be improved by considering the walking direction and position of users to further improve performance.

6.4 Limitation and Future Work

Although NEPR and Hm served as practical substitutes for conventional resets and showed overall improvements over TiP by supporting continuous forward walking, improving task performance, and enhancing the user experience, several limitations remain. First, the operating logic must be generalized beyond the current obstacle-free assumption. Integrating real-time path planning will allow NEPR to navigate irregular geometries and avoid dynamic obstacles in complex physical spaces. These changes will extend the usage patterns observed in Fig. 8 to less regular spaces. Portal activation requires a discernible navigation target (PoI). Intent inference via gaze, dwell time, or goal prediction [16] could increase robustness in exploratory scenarios. Optional user-initiated activation or veto could improve control and comfort. Portal transitions introduce discrete spatial updates that may affect spatial updating and memory. Visual binding between the primary and secondary worlds, gentle and well-cued transitions, and approximately matching physical walking distance to virtual progress may help maintain spatial awareness. While most participants reported no discomfort with the visual design of the secondary world, further research is needed to determine the optimal design of the virtual corridor for a wider variety of environments. Since NEPR/Hm moves users toward PoIs, while TiP does not, TWD and ET primarily capture end-to-end task performance, rather than the isolated cost of a reset. Evaluations against portal-style baselines under matched advancement budgets or a non-advancing NEPR variant would refine interpretation. Combining portal resets with various types of redirection [4, 12, 50] and testing generalization are promising directions.

7 CONCLUSION

We introduced the NEPR, which is designed to prevent frequent resets for distant targets and stop users from having to turn around while walking using the traditional turn-in-place method. The NEPR repositions and reorients users without interrupting their gait. This allows users to make continuous progress toward distant targets, improving task performance and perceived usability, as demonstrated in our user studies. For near-range targets, the turn-in-place reset remains efficient. Therefore, a hybrid approach that applies the NEPR to distant targets and the turn-in-place reset to near-range targets balances this tradeoff effectively. Further research on the new reset method could extend its applicability to irregular environments with physical obstacles and refine the strategy for selecting resets to account for the user's state and walking pattern. Taken together, these results suggest that incorporating resets into forward locomotion can counteract the effects of spatial constraints. Thus, NEPR emerges as a practical alternative to turn-in-place resets for redirected walking in many scenarios.

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